

Technology Stack

Date	5-7-2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies used to develop the EduGenie AI Learning Assistant, including the programming languages, frameworks, frontend and backend tools, database, AI APIs, and supporting software. A suitable technology stack ensures that the application is efficient, maintainable, scalable, and capable of delivering AI-powered educational features such as question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations. Fill out the table below with the technologies used in each architectural component and provide a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2	Creates a responsive web interface and dynamically displays AI-generated content to users.
2	Backend / Server-Side	Python, FastAPI	Handles application logic, processes user requests, and communicates with AI services efficiently.
3	Database / Data Storage	SQLite, JSON Files	Stores application data, user information, quiz results, and learning records in a lightweight format.
4	Version Control & Development	GitHub	Tracks source code changes, supports collaboration, and maintains project versions.
5	Hosting / Deployment	Uvicorn (Local), Google Cloud Run, GitHub Pages	Runs the FastAPI application and deploys the frontend and backend for public access.
6	Third-Party APIs / AI Tools	Google Gemini 2.5 Flash API	Provides AI-powered question answering, explanations, quizzes, summaries, and personalized learning paths.